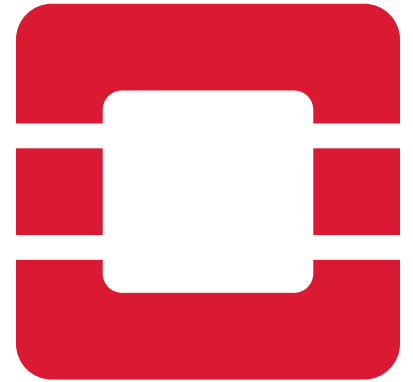


# OpenStack



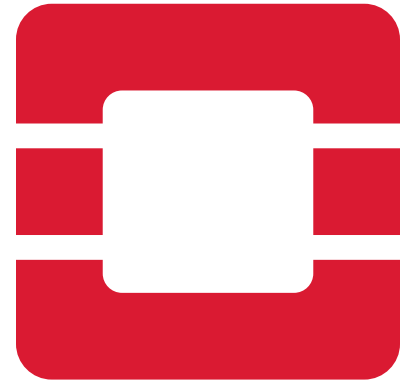
# Targets

- Deployment
- Operations
- Networking
- Monitoring
- Troubleshooting
- Automation
- Testing

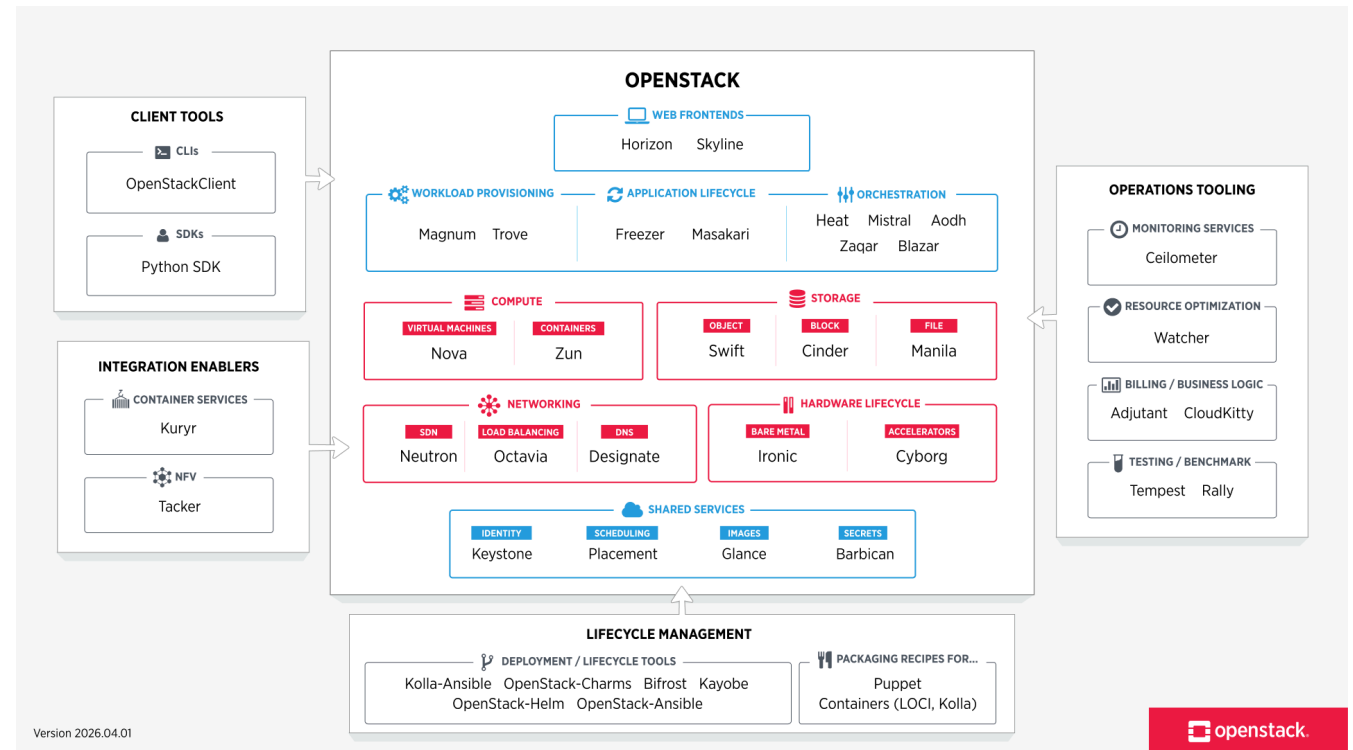
# OpenStack Distributions

## Catalog

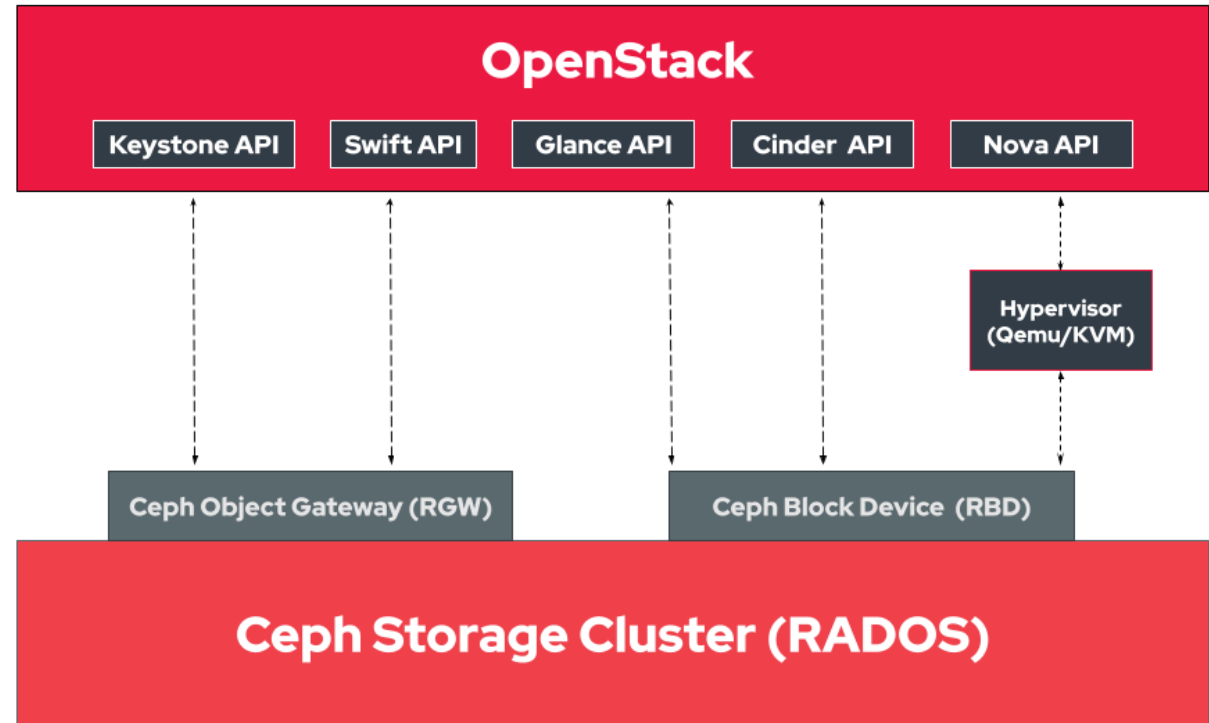
- OSISM
- Mirantis
- Canonical
- RedHat
- FishOS



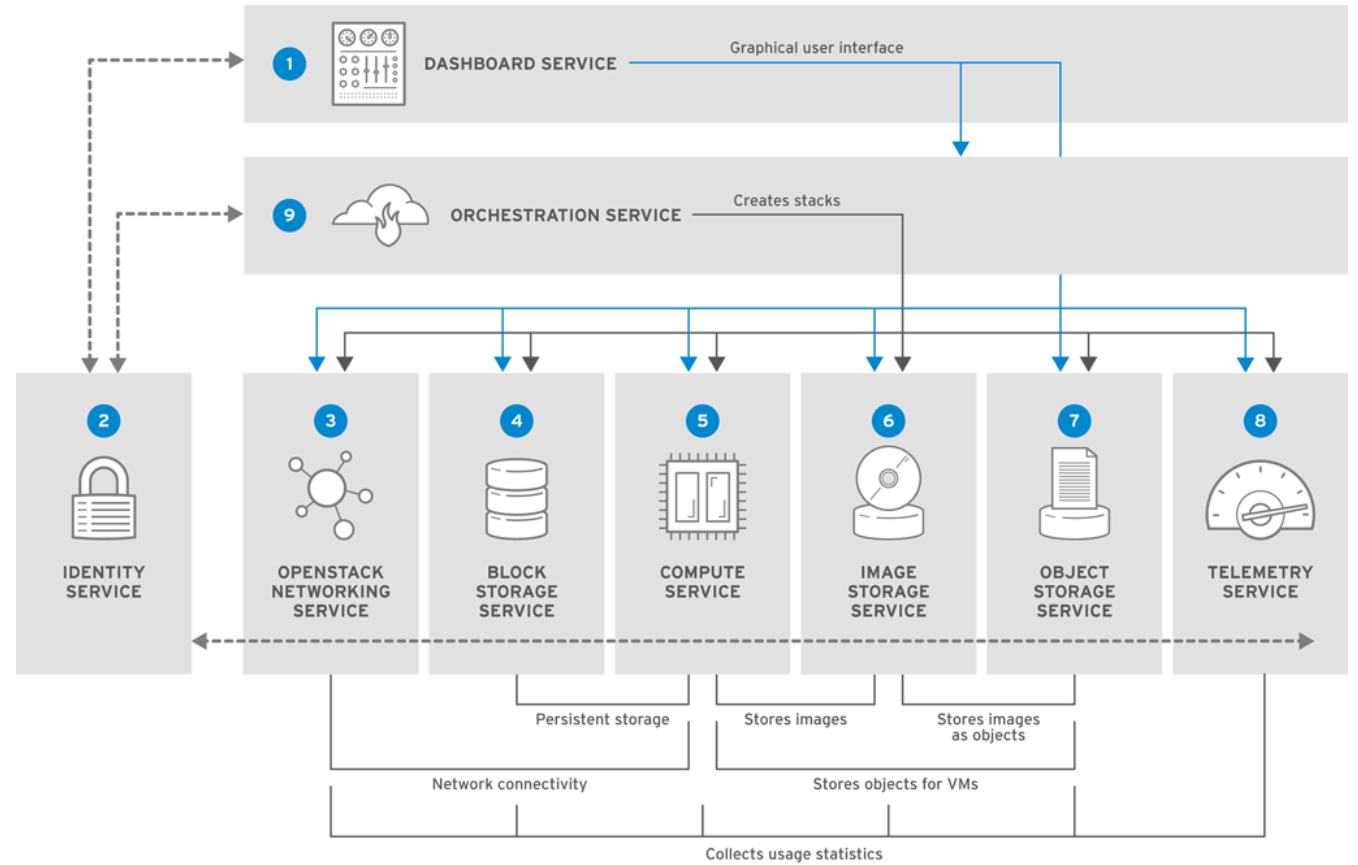
# Architecture



# Architecture

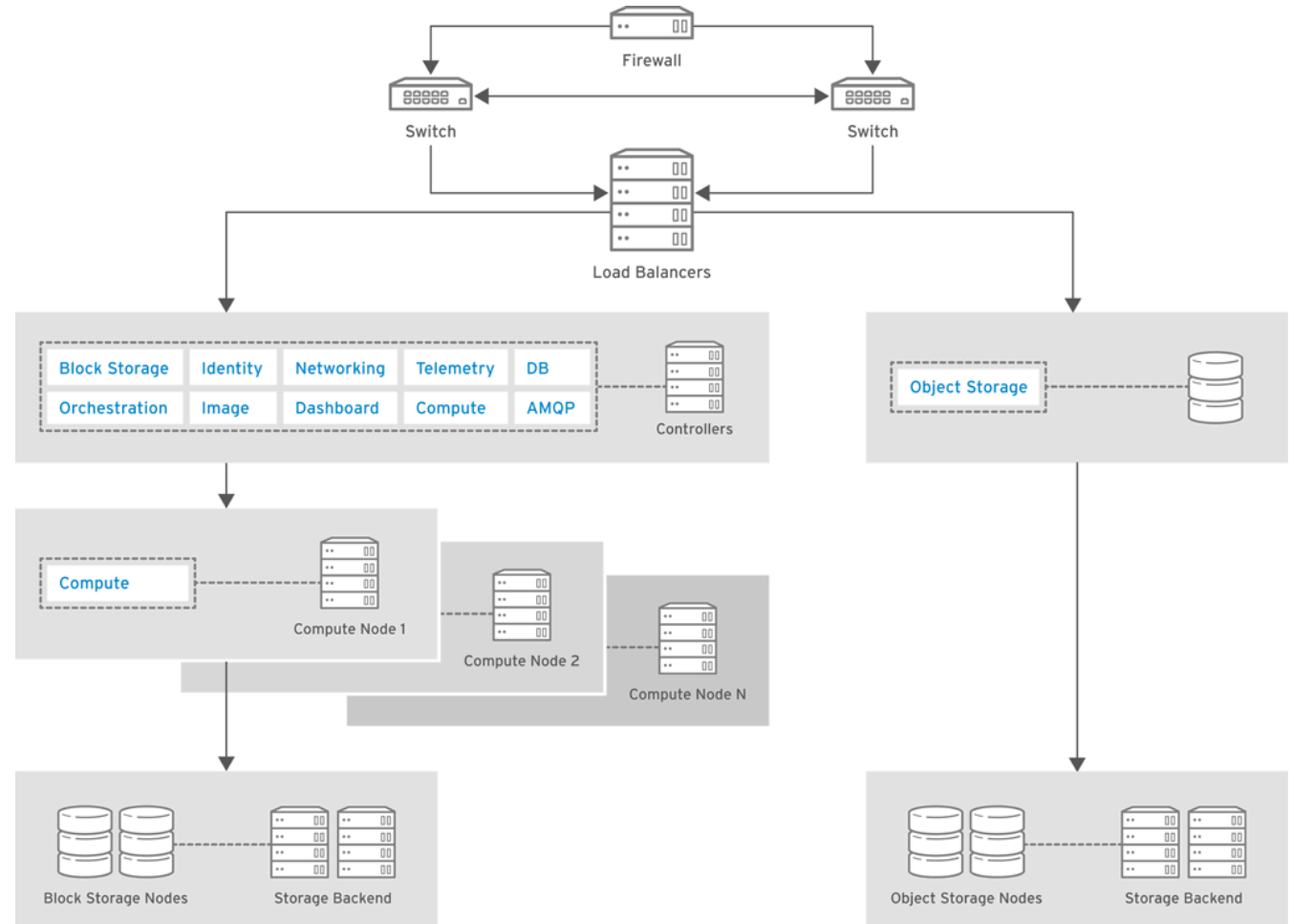


# Architecture



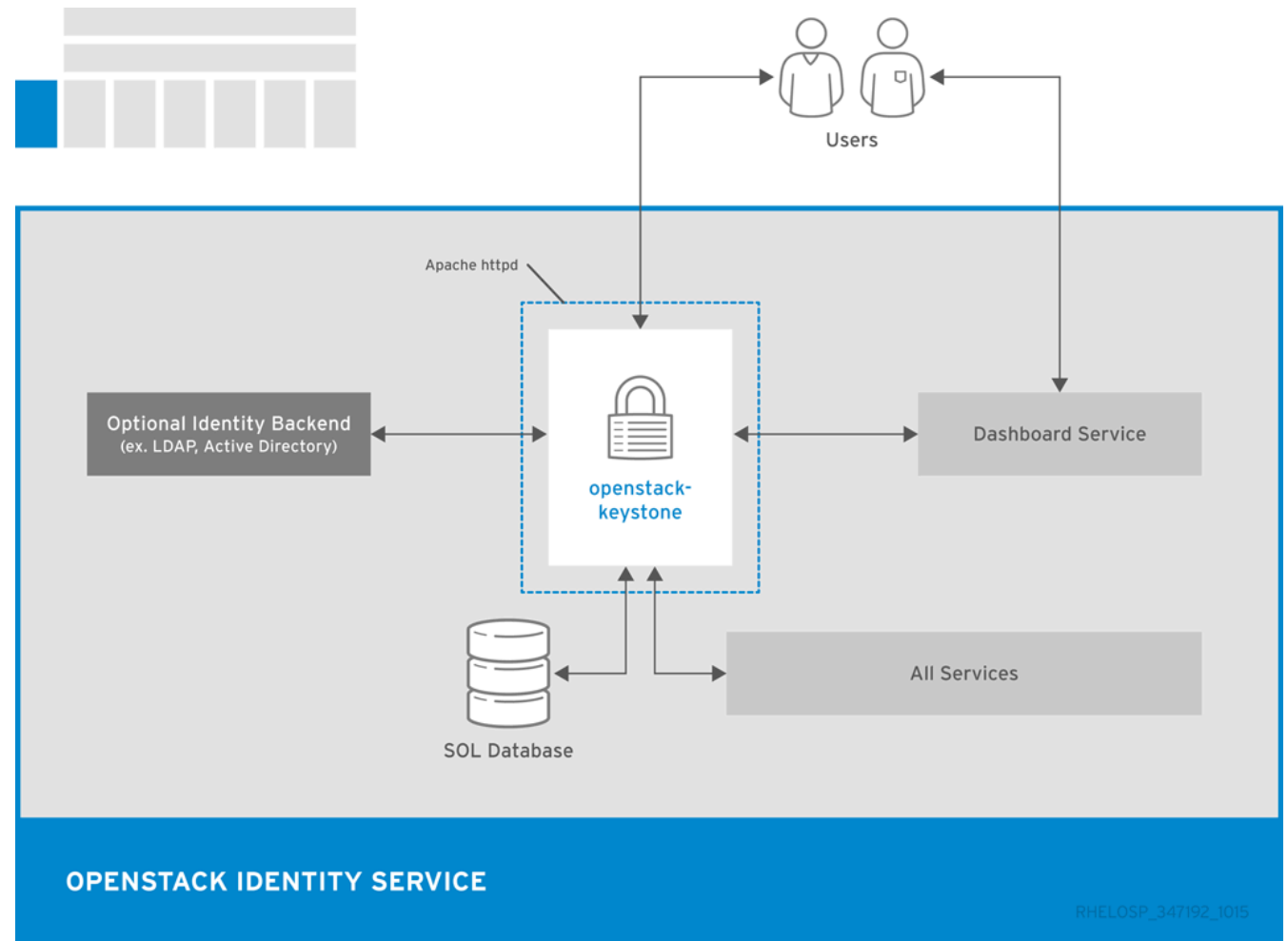
RHELOSP\_347192\_1015

# Architecture

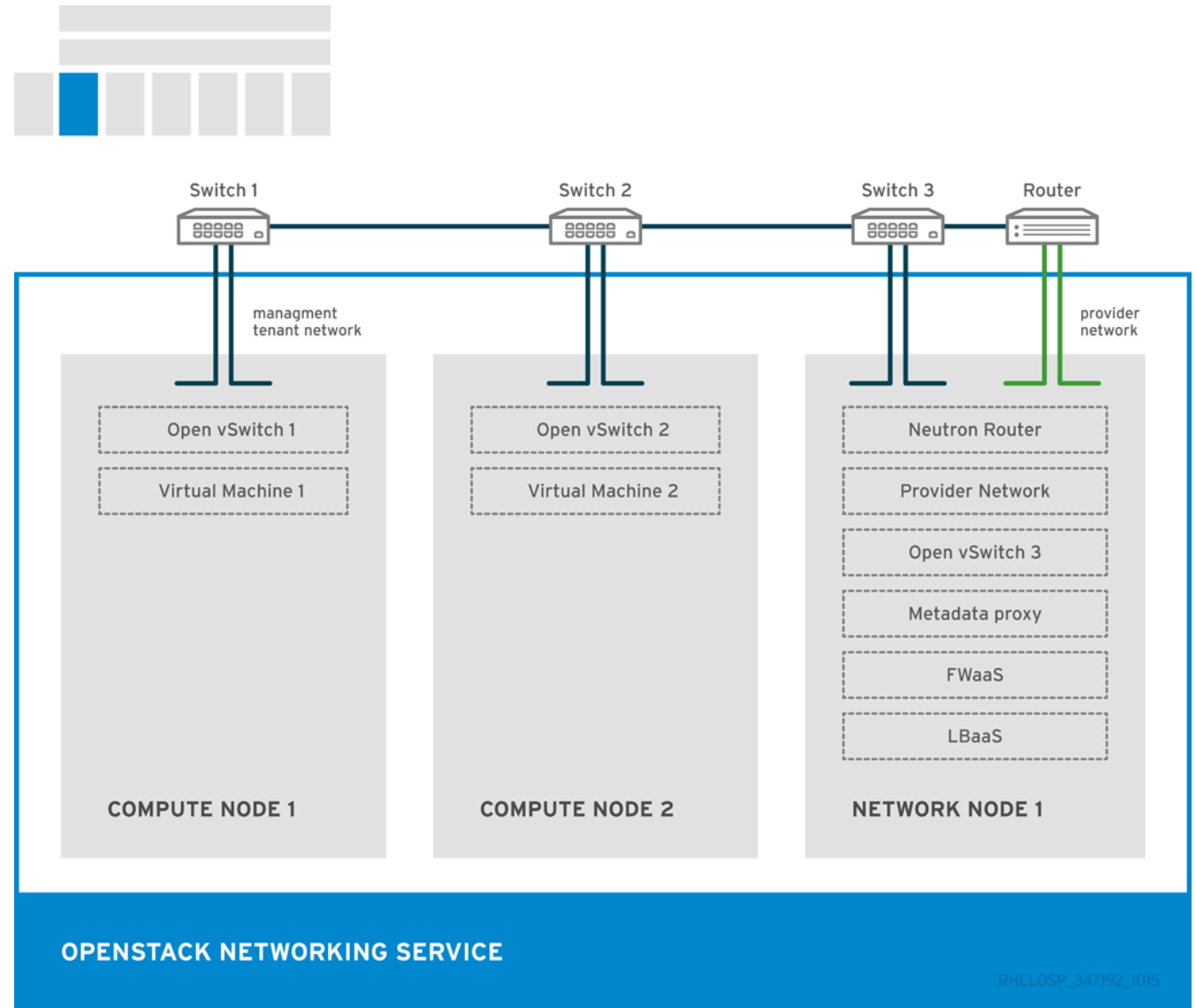


RHELOSP\_347192\_1015

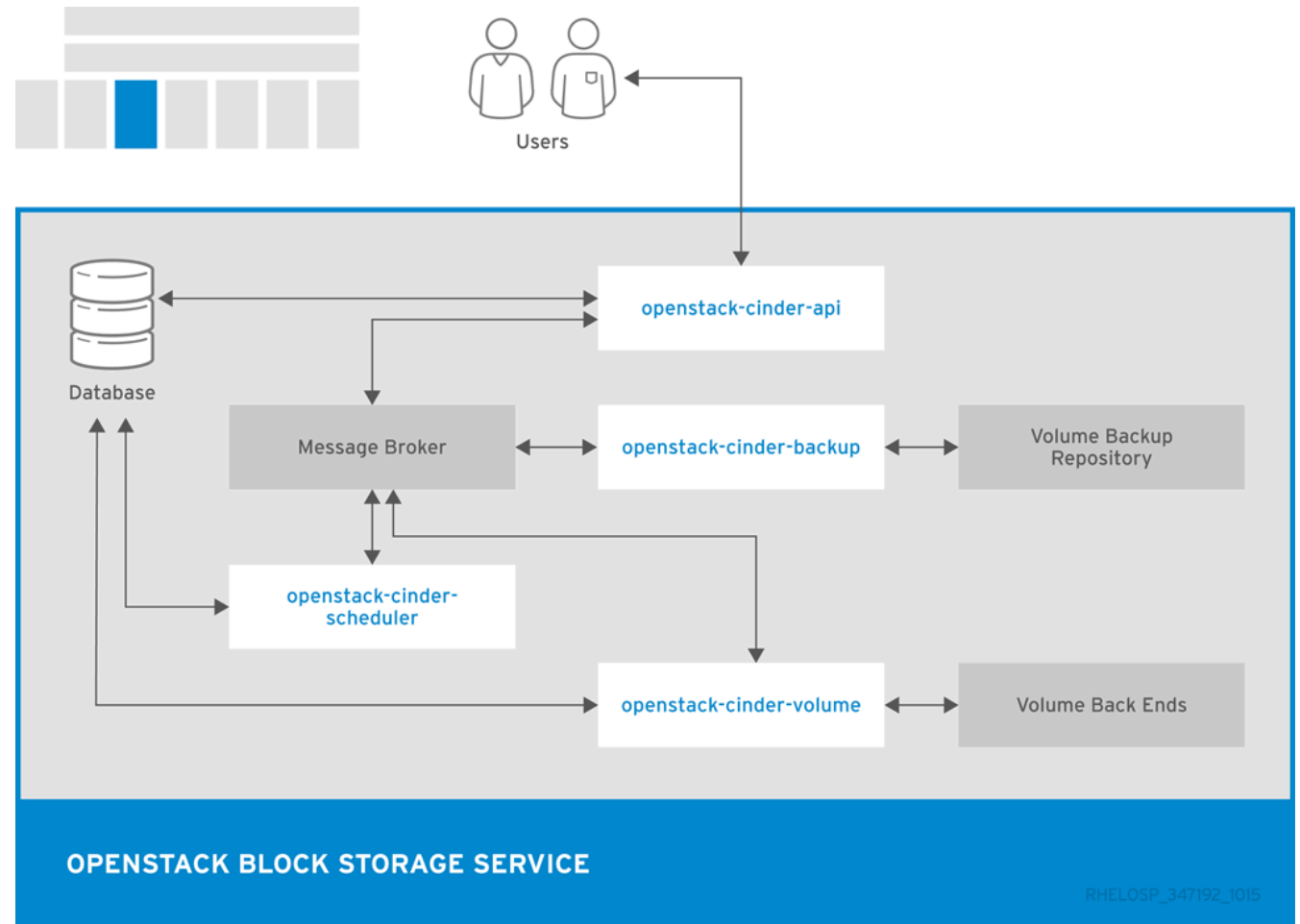
# Architecture



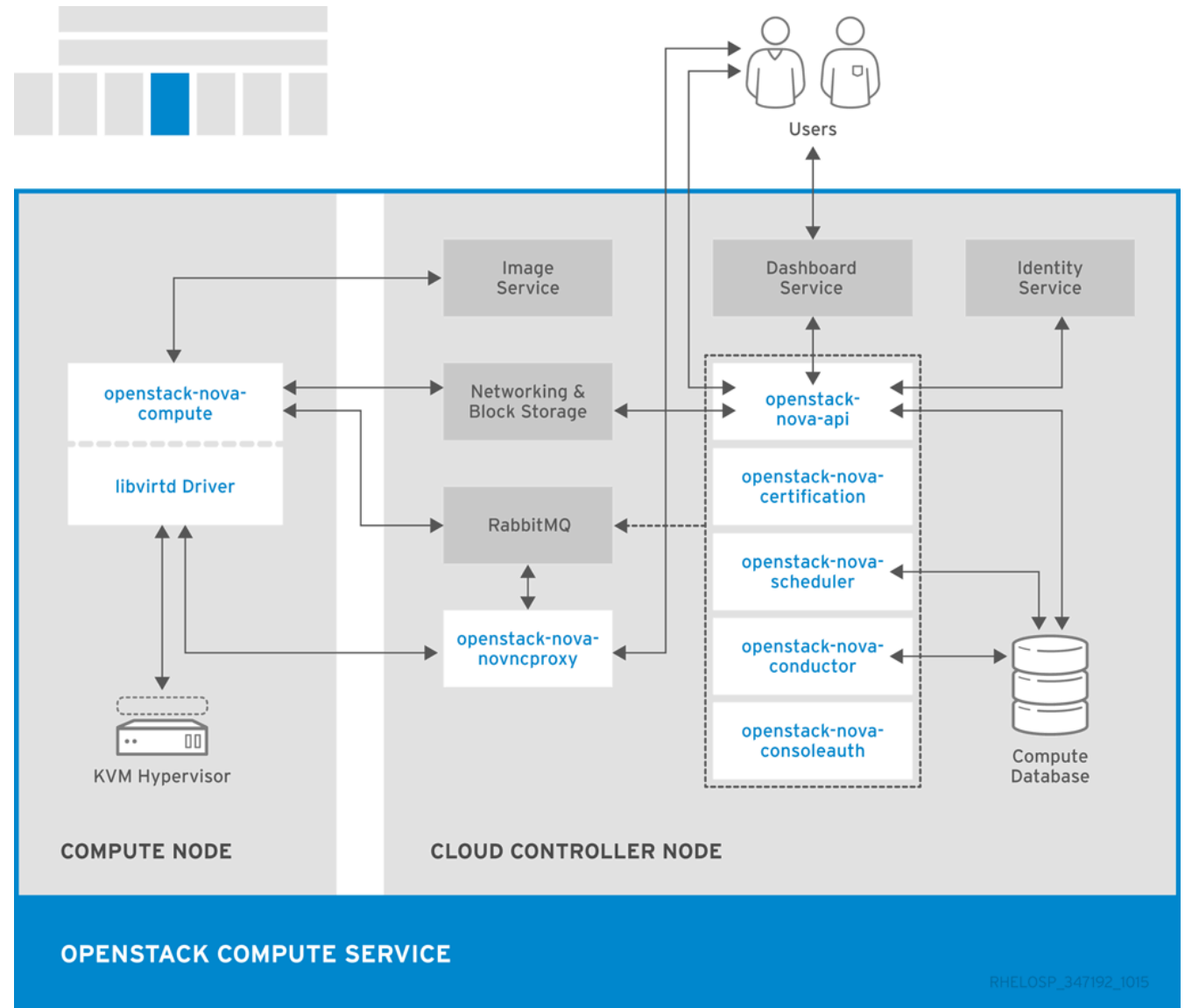
# Architecture



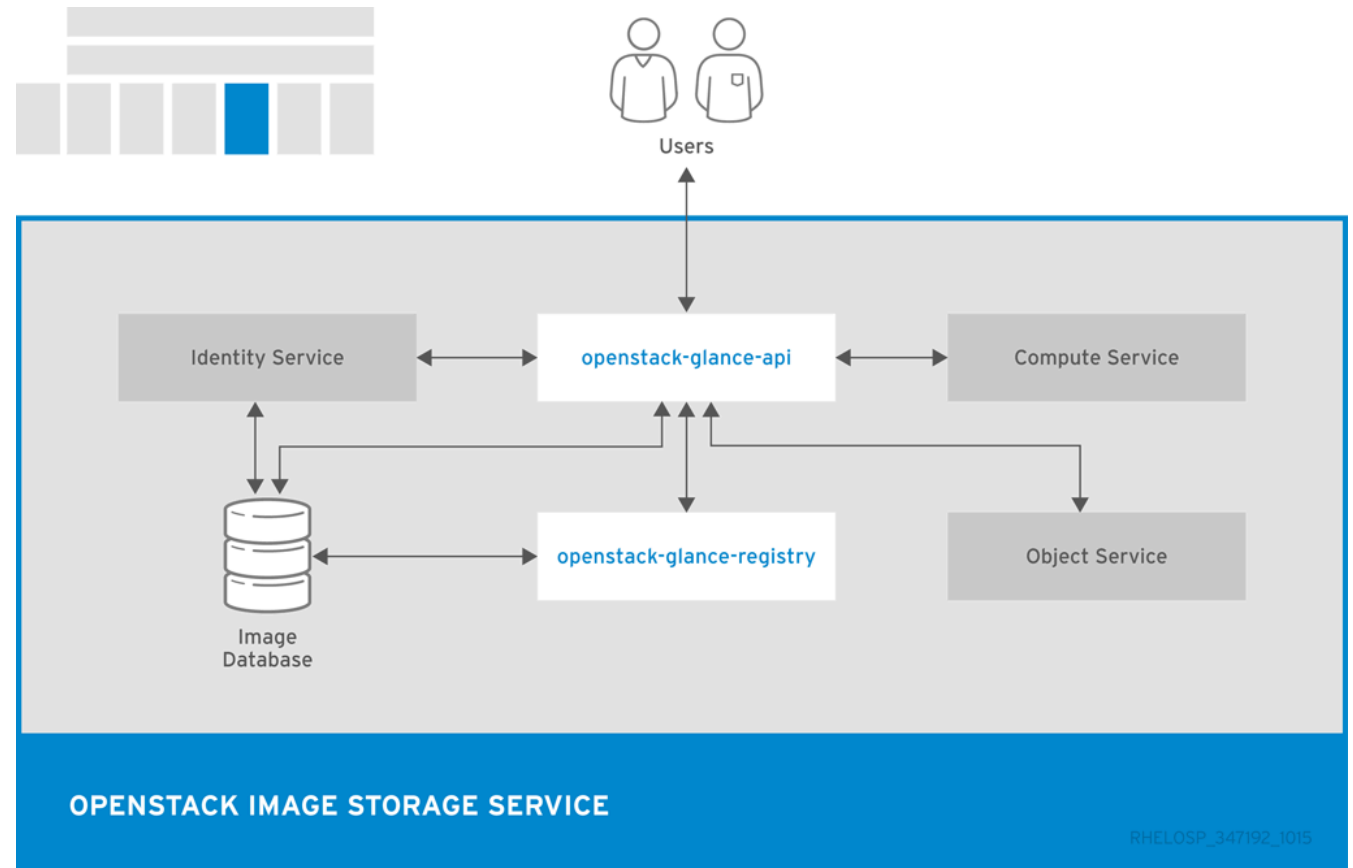
# Architecture



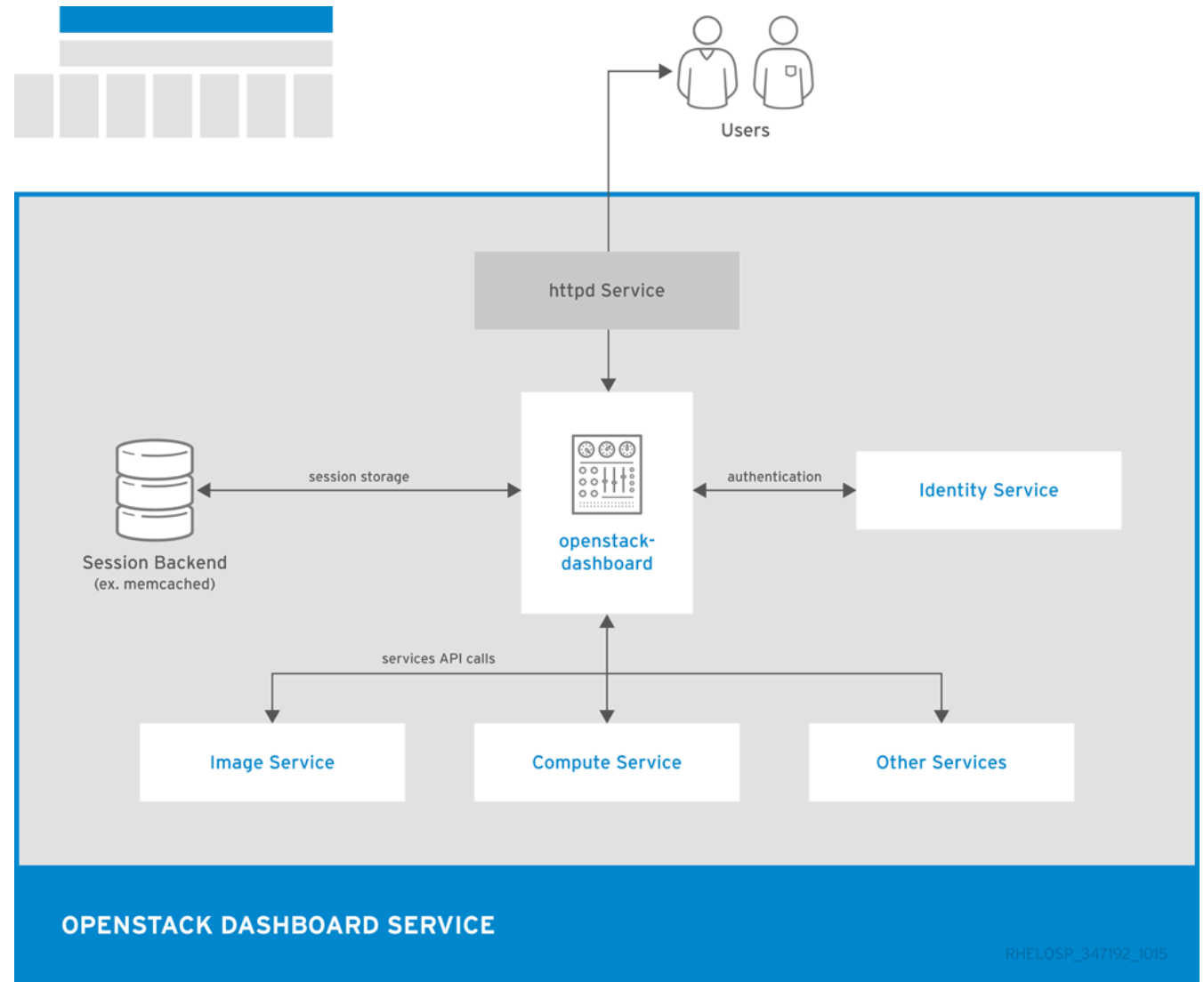
# Architecture



# Architecture

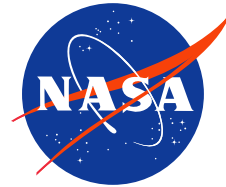


# Architecture



# Users

Walmart 



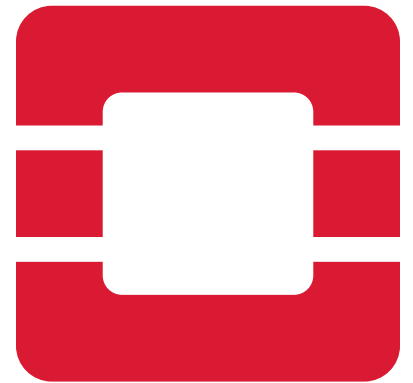
**Bloomberg**

# Contributors



# Deployment Approaches

- openstack-ansible
- kolla-ansible
- OpenStack on OpenShift
- OpenStack-Helm
- OSISM
- YAOOK
- ...



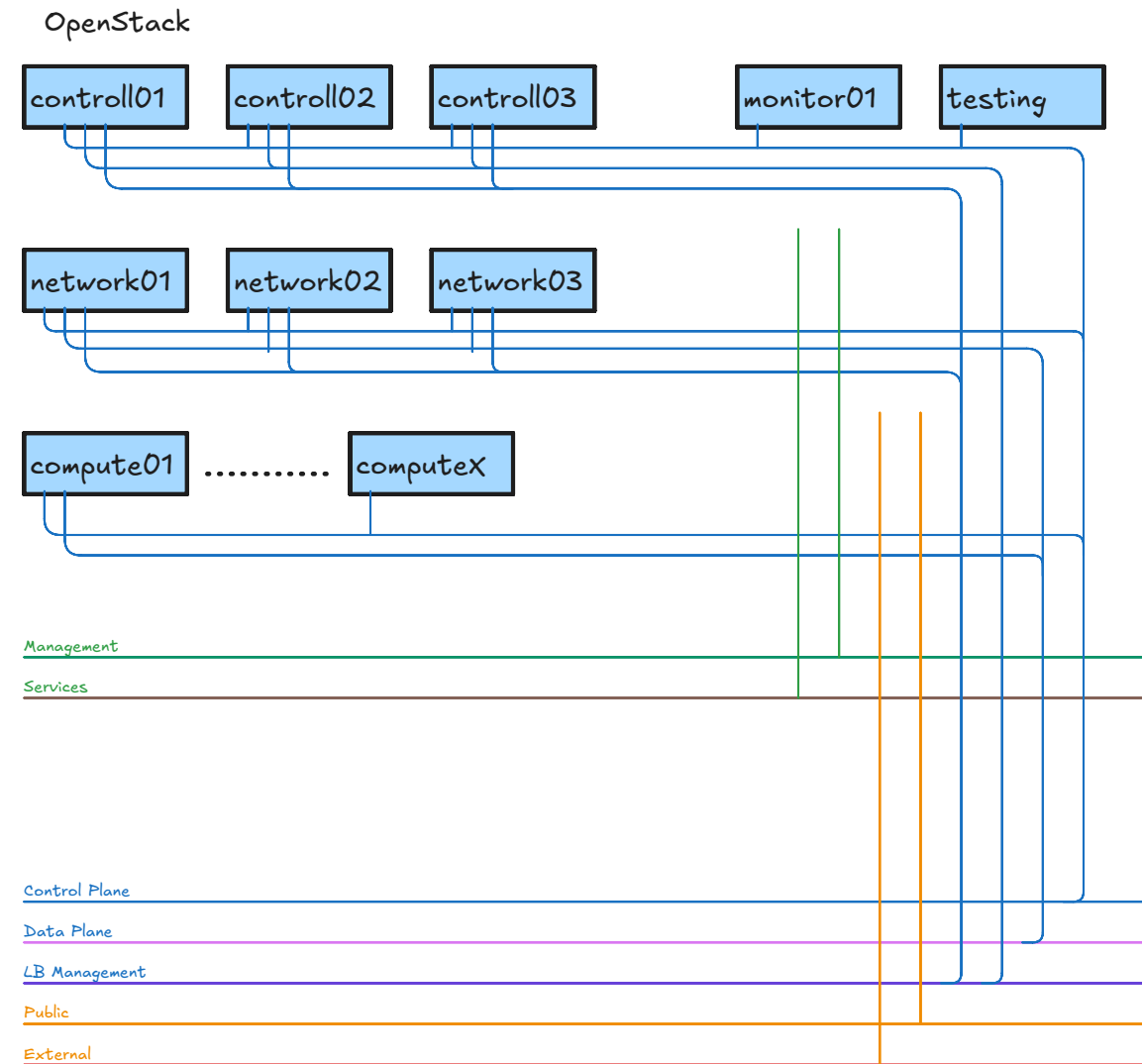
# Deploy OpenStack



# KOLLA

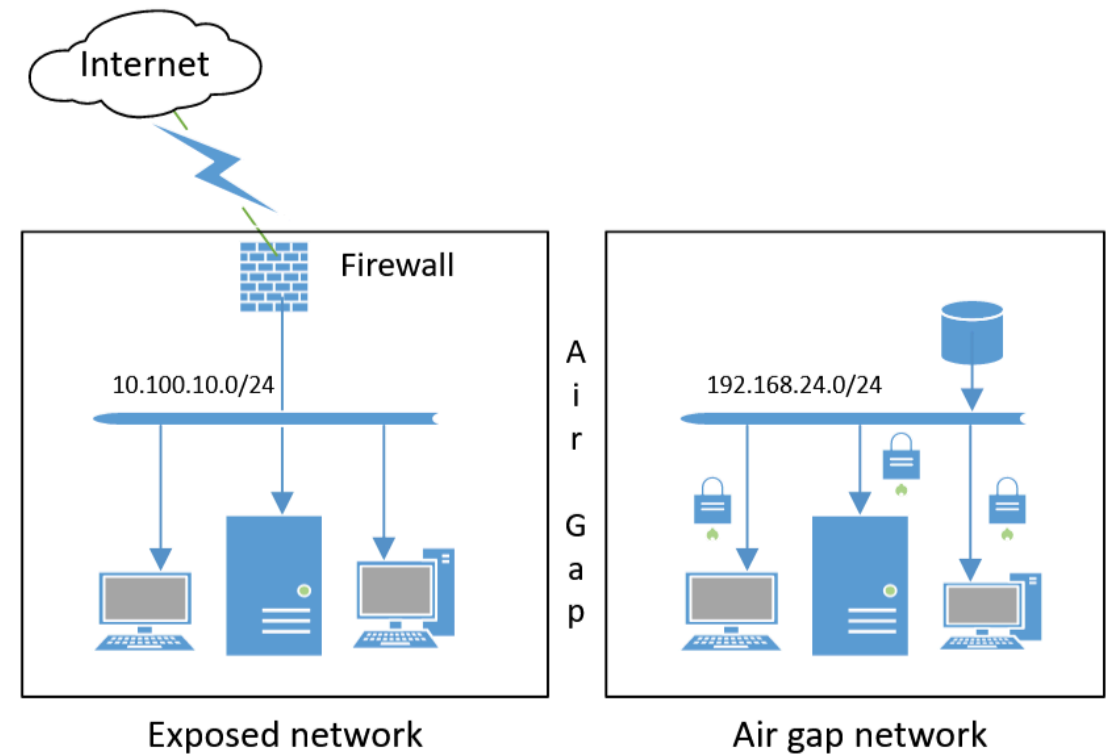
*an OpenStack Community Project*

# OpenStack Network



# Environment Network

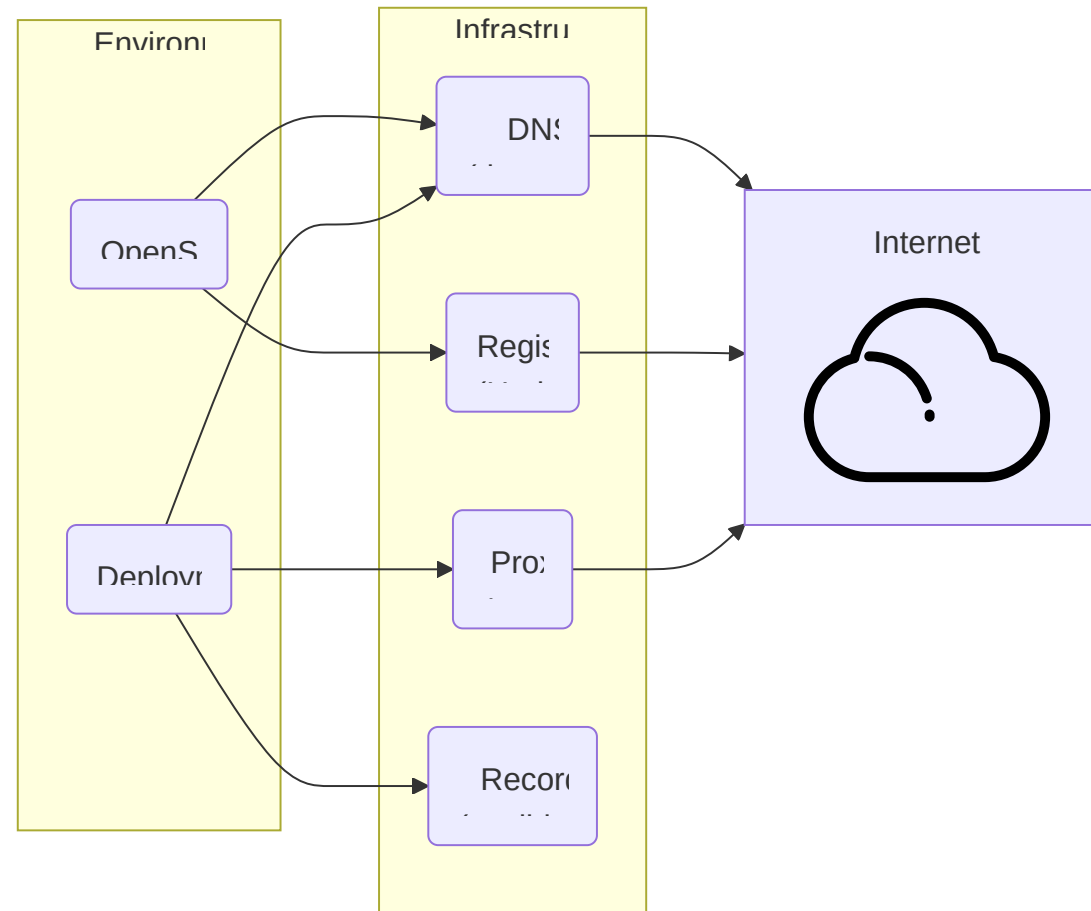
# Air-Gapped Environment



# Workshop Hardware and VMs

- Hardware: 60 vCPUs, 160 GB RAM, 1 TB disk
- Deployment VM
- Infra VMs: registry, proxy, dns, recorder
- OpenStack VMs: 3x Controller, 3x Network, 3(+ )x Compute, 1xMonitoring, 1xTesting

# Infrastructure



# Deployment Node

- Python virtual envs
- kolla-ansible

# Deploy OpenStack

```
# Prepare Python virtual environment
cd ~/openstack/

python3 -m venv $VENV_PATH

ln -s $VENV_PATH ~/venv/openstack
source $VENV_PATH/bin/activate

# Install needed pip packages
pip install -U pip
pip install -r requirements.txt

# Install dependencies for kolla-ansible
kolla-ansible install-deps

# ⚠ NOTE: fix for prometheus 05.2026
# pip install "bcrypt<5.0.0"

# Install additional dependencies
gilt overlay

# Reload environment
source ~/.profile
```



# Deploy OpenStack

```
# Prepare ansible inventory

# Copy inventory file
cp ~/venv/openstack/share/kolla-ansible/ansible/inventory/multinode inventory/

# Patch inventory file and review inventory/wrx/ directory afterwards
crudini --del inventory/multinode control
crudini --del inventory/multinode network
crudini --del inventory/multinode compute
crudini --del inventory/multinode monitoring
crudini --del inventory/multinode storage
crudini --del inventory/multinode loadbalancer:children

# Prepare passwords
cp ~/venv/openstack/share/kolla-ansible/etc_examples/kolla/passwords.yml \
  custom-config/wrx/passwords.yml

kolla-genpwd -p custom-config/wrx/passwords.yml
```



# KOLLA

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# Deploy OpenStack

# Overview of kolla-ansible commands



```
kolla-ansible bootstrap-servers -i inventory/wrx
kolla-ansible octavia-certificates -i inventory/wrx
kolla-ansible prechecks -i inventory/wrx
kolla-ansible pull -i inventory/wrx
kolla-ansible deploy -i inventory/wrx -vvvv
kolla-ansible post-deploy -i inventory/wrx
kolla-ansible check -i inventory/wrx
kolla-ansible destroy -i inventory/wrx <--yes-i-really-really-mean-it>
kolla-ansible mariadb_recovery -i inventory/wrx
kolla-ansible reconfigure -i inventory/wrx
kolla-ansible upgrade -i inventory/wrx
kolla-ansible stop -i inventory/wrx
kolla-ansible deploy-containers -i inventory/wrx
kolla-ansible prune-images -i inventory/wrx
```



# KOLLA

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# Deploy OpenStack

```
# Prepare the kolla-ansible deployment
```



```
kolla-ansible bootstrap-servers    -i inventory/wrx  
kolla-ansible octavia-certificates -i inventory/wrx  
kolla-ansible prechecks           -i inventory/wrx
```

```
# ⚠ NOTE: this will take time  
kolla-ansible pull                 -i inventory/wrx
```



# Deploy OpenStack

```
# Prepare the kolla-ansible deployment

# ⚠ NOTE: this will take time
kolla-ansible deploy          -i inventory/wrx

# Post deployment steps
kolla-ansible post-deploy    -i inventory/wrx
kolla-ansible check          -i inventory/wrx
```



# Bootstrap a Test Environment

```
# Create first basic test resources
KOLLA_CONFIG_PATH=~/.openstack/custom-config/wrx/ \
ENABLE_EXT_NET=1 \
EXT_NET_CIDR=10.30.0.0/24 \
EXT_NET_GATEWAY=10.30.0.1 \
EXT_NET_RANGE="start=10.30.0.240,end=10.30.0.245" \
DEMO_NET_DNS=10.30.0.1 \
~/venv/openstack/share/kolla-ansible/init-runonce
```



# KOLLA

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# Access

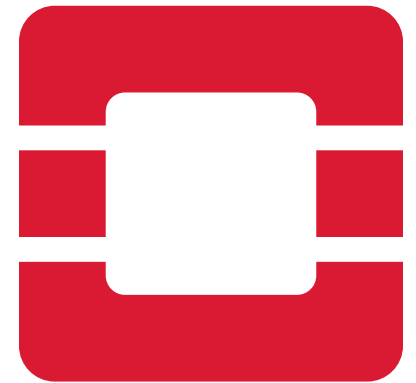
Horizon <http://int.os.wrx.sckt.net>

Skyline <http://int.os.wrx.sckt.net:9999>

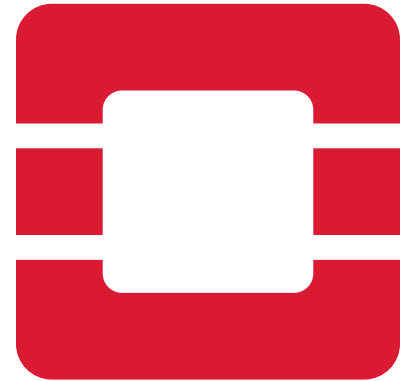
CLI Command Line Interface

Check the following files to use with CLI:

- custom-config/wrx/admin-openrc.sh
- custom-config/wrx/clouds.yaml



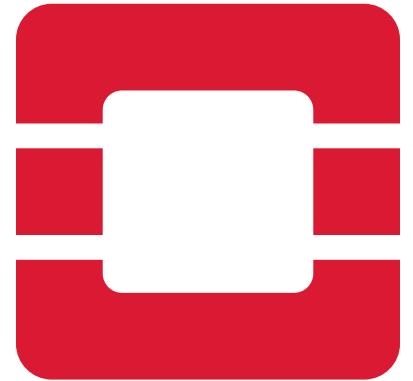
# Operations



# Service Overview

```
# Review the OpenStack environment
openstack service list
openstack endpoint list
openstack catalog list

openstack compute service list
openstack volume service list
openstack network agent list
openstack orchestration service list
```

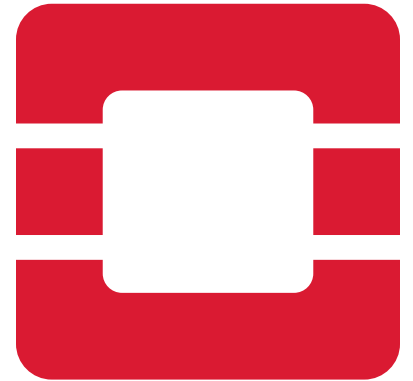


# Replace a Network Node

```
# 🔍 Review data stored on the node. Is there anything to back up?
docker volume ls
du -lhs /var/lib/docker/volumes/*

# 💥 Break the network03 node
apt remove docker* -y
rm -rf /etc/kolla /var/log/kolla \
      /etc/systemd/system/kolla* \
      /etc/systemd/system/multi-user.target.wants/kolla-* \
      /etc/docker/daemon.json
umount /var/lib/docker; wipefs /dev/sdb -a; mkfs.ext4 /dev/sdb
reboot

# 🚚 Redeploy OpenStack services on network03 from the deployment node
kolla-ansible bootstrap-servers -i inventory/wrx --limit ~network03
kolla-ansible prechecks        -i inventory/wrx --limit ~network03
kolla-ansible pull              -i inventory/wrx --limit ~network03
kolla-ansible reconfigure       -i inventory/wrx --limit ~network03
```



# Replace a Controller Node

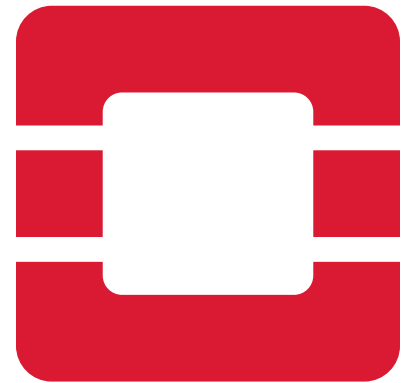
```
# 🔍 Review data stored on the node. Is there anything to back up?
docker volume ls
du -lhs /var/lib/docker/volumes/*

# ⚡ Break the controller03 node
systemctl stop kolla-* docker.service docker.socket
apt remove docker* -y
rm -rf /etc/kolla /var/log/kolla \
      /etc/systemd/system/kolla* \
      /etc/systemd/system/multi-user.target.wants/kolla-* \
      /etc/docker/daemon.json
umount /var/lib/docker; wipefs /dev/sdb -a; mkfs.ext4 /dev/sdb
reboot

# 🚚 Redeploy OpenStack services on controller03 from the deployment node
kolla-ansible bootstrap-servers -i inventory/wrx --limit ~controller03
kolla-ansible prechecks        -i inventory/wrx --limit ~controller03
kolla-ansible pull              -i inventory/wrx --limit ~controller03

# 🍲 this will take a while
kolla-ansible reconfigure       -i inventory/wrx --limit ~controller03 --skip-tags keystone

# ⚠️ we have to reconfigure keystone across all controller nodes ⚠️
kolla-ansible reconfigure       -i inventory/wrx --tags keystone
```



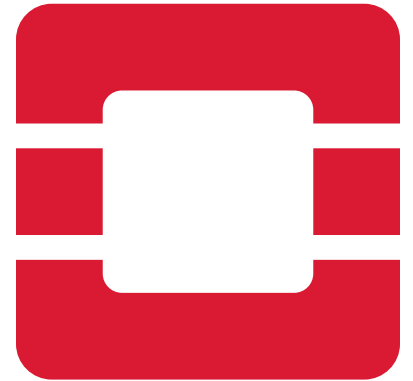
# Replace a Compute Node

```
# 🔍 Review data stored on the node. Is there anything to back up?
docker volume ls
du -lhs /var/lib/docker/volumes/*
docker exec -ti nova_libvirt virsh list --all

# 🛑 Drain the OpenStack compute host
openstack server list --all --host <COMPUTE_NAME>
openstack compute service set --disable <COMPUTE_NAME> nova-compute
for vm in $(openstack server list --host <COMPUTE_NAME> --all-projects -c ID -f value);
do
    echo "Evacuating VM: $vm"
    openstack server migrate --live $vm
done
openstack server list --all --host <COMPUTE_NAME>

# 💥 Break the node
# On the node
systemctl stop kolla-* docker.service docker.socket
apt remove docker* -y
rm -rf /etc/kolla /var/log/kolla \
    /etc/systemd/system/kolla* \
    /etc/systemd/system/multi-user.target.wants/kolla-* \
    /etc/docker/daemon.json
cp -pr /var/lib/docker/volumes ~/
umount /var/lib/docker; wipefs /dev/sdb -a; mkfs.ext4 /dev/sdb
reboot

# 🚚 Redeploy OpenStack services on compute03 from the deployment node
kolla-ansible bootstrap-servers -i inventory/wrx --limit ~<COMPUTE_NAME>
kolla-ansible prechecks -i inventory/wrx --limit ~<COMPUTE_NAME>
kolla-ansible pull -i inventory/wrx --limit ~<COMPUTE_NAME>
kolla-ansible reconfigure -i inventory/wrx --limit ~<COMPUTE_NAME>
```



# Debugging

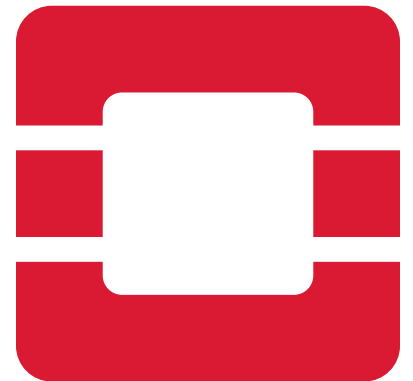
```
# 🐞 Check
openstack compute service list

# 🔍 Take a look at logs on compute03
tail -f /var/log/kolla/nova/nova-compute.log

# 🔍 Compare files on nova_compute volume and in the backup
ls -lh ~/volumes/nova_compute/_data/ /var/lib/docker/volumes/nova_compute/_data/

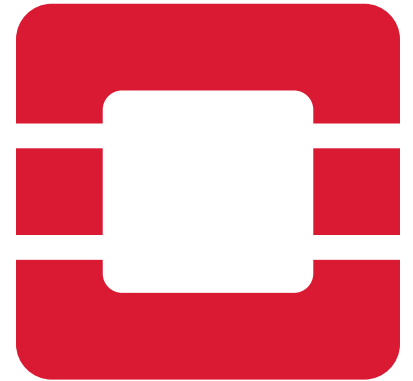
# ⚙️ Repair
cp ~/volumes/nova_compute/_data/compute_id /var/lib/docker/volumes/nova_compute/_data/

# 🐞 Check the status again and test
openstack compute service list
openstack compute service set --enable compute03 nova-compute
openstack server migrate --live --host compute03 <SERVER_ID>
openstack server list --all --long
```



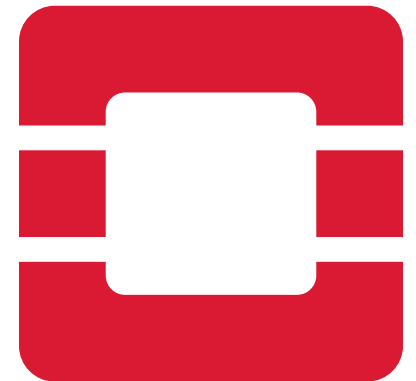
# Scale-out

- How to scale out controllers?
- How to scale out network nodes?
- How to scale out compute nodes?

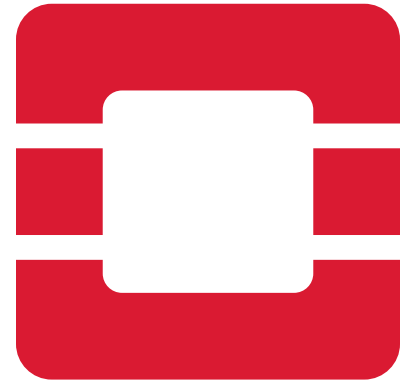


# Add a New Compute Host

```
# 📝 Modify inventory, put the new compute host into the hypervisor section 📋  
vim inventory/wrx/20_openstack  
  
# 🔄 Check the new host  
  
# 🚚 Deploy OpenStack services on the new compute host  
kolla-ansible bootstrap-servers -i inventory/wrx --limit ~compute12  
kolla-ansible prechecks        -i inventory/wrx --limit ~compute12  
kolla-ansible pull              -i inventory/wrx --limit ~compute12  
kolla-ansible deploy            -i inventory/wrx --limit ~compute12  
  
# 🔄 Check the new host  
openstack compute service list  
openstack server migrate --live --host compute12 <SERVER_ID>  
  
docker exec -ti nova_libvirt virsh list --all # on the host
```

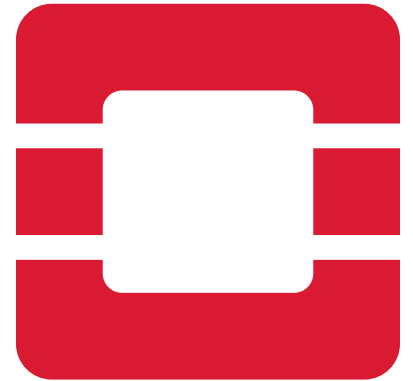


# Remove a Compute Host



# Reboot an OpenStack Host

- How to reboot controllers?
- How to reboot network nodes?
- How to reboot compute nodes?



# Debugging RabbitMQ

# Some commands to debug RabbitMQ errors

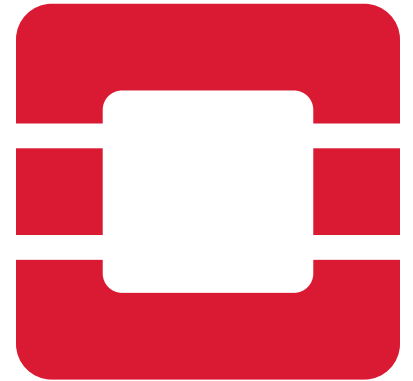


```
rabbitmqctl status  
rabbitmqctl cluster_status
```

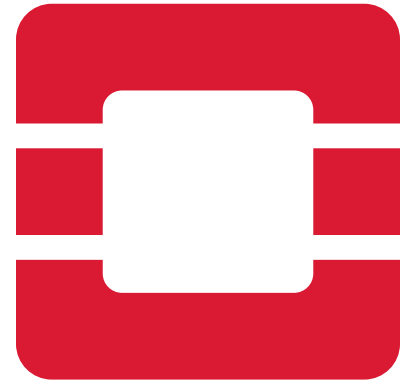
```
rabbitmq-plugins list  
rabbitmq-plugins disable <PLUGIN>  
rabbitmq-plugins enable <PLUGIN>
```

```
rabbitmqctl list_queues name, messages, messages_ready, messages_unacknowledged  
rabbitmqctl list_bindings
```

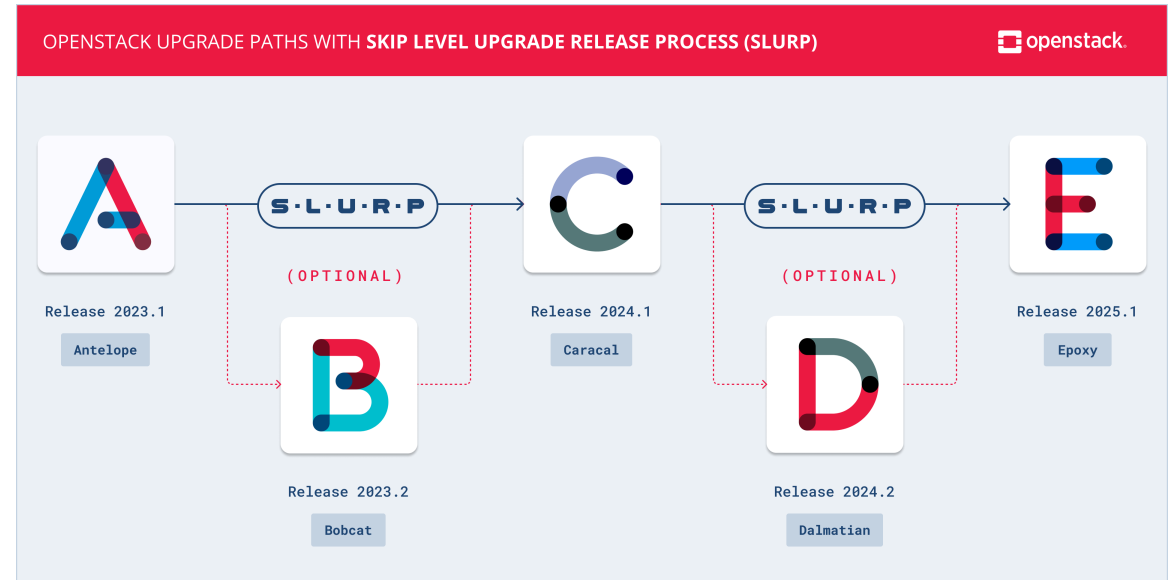
```
rabbitmqctl purge_queue <QUEUE_NAME>  
rabbitmqctl delete_queue <QUEUE_NAME>
```



# Upgrade



# SLURP Model

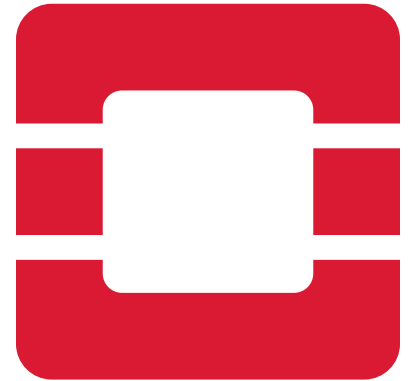


# Roll out the Next Release

```
# 📝 Update kolla-ansible
CUR_KA_VER=$(pip list | grep kolla | awk '{print $NF}' | awk -F . '{print $1}')

echo $VIRTUAL_ENV
deactivate
python3 -m venv ~/venv/kolla-ansible-$((CUR_KA_VER+1))
source ~/venv/kolla-ansible-$((CUR_KA_VER+1))/bin/activate

pip install -U pip
pip install -r requirements.txt
pip install -U "kolla-ansible<$((CUR_KA_VER + 2)).0"
kolla-ansible install-deps
# ⚠️ NOTE: fix for prometheus 05.2026
# pip install "bcrypt<5.0.0"
```

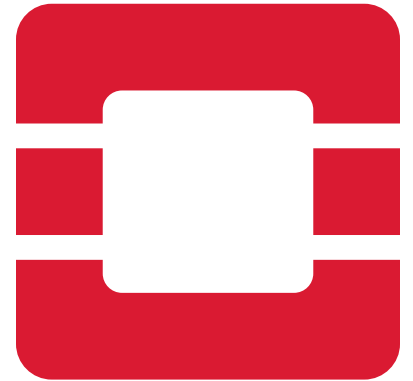


# Roll out the Next Release

## Inventory

```
# Update inventory
mv inventory/multinode inventory/multinode.old
cp $VIRTUAL_ENV/share/kolla-ansible/ansible/inventory/multinode inventory/
crudini --del inventory/multinode control
crudini --del inventory/multinode network
crudini --del inventory/multinode compute
crudini --del inventory/multinode monitoring
crudini --del inventory/multinode storage
crudini --del inventory/multinode loadbalancer:children

# Review the differences
diff -y --suppress-common-lines inventory/multinode inventory/multinode.old
```

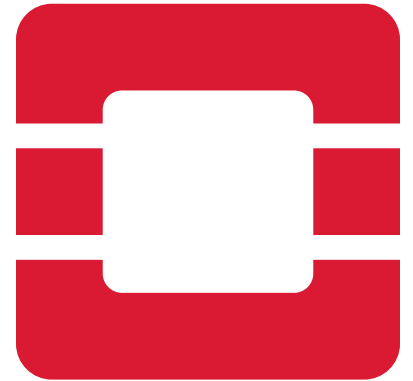


# Roll out the Next Release

## Passwords

```
# Update passwords
mv custom-config/wrx/passwords.yml custom-config/wrx/passwords.yml.old
cp $VIRTUAL_ENV/share/kolla-ansible/etc_examples/kolla/passwords.yml \
  custom-config/wrx/passwords.yml
kolla-genpwd -p custom-config/wrx/passwords.yml
kolla-mergepwd \
  --old custom-config/wrx/passwords.yml.old \
  --new custom-config/wrx/passwords.yml \
  --final custom-config/wrx/passwords.yml

# Review the differences
diff -y --suppress-common-lines custom-config/wrx/passwords.yml custom-config/wrx/passwords.yml.old
```



# Roll out the Next Release

## Go

```
# 📁 Create config backups
ansible all -b -m ansible.builtin.shell -a \
  "mkdir -p ~/backup; cp -r /etc/kolla ~/backup/$(date +%y%m%d%H%M%S)_kolla"

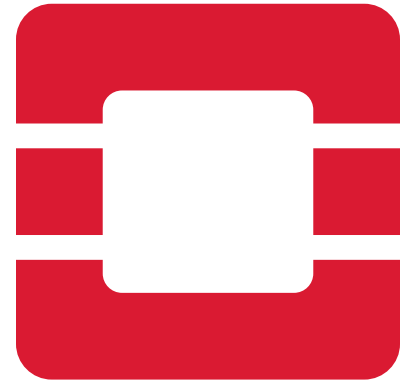
# 📁 Force to collect facts
kolla-ansible gather-facts -i inventory/wrx -e "ansible_python_interpreter=/usr/bin/python3"

# ansible all -m ansible.builtin.setup

# 🔄 Check
kolla-ansible prechecks -i inventory/wrx

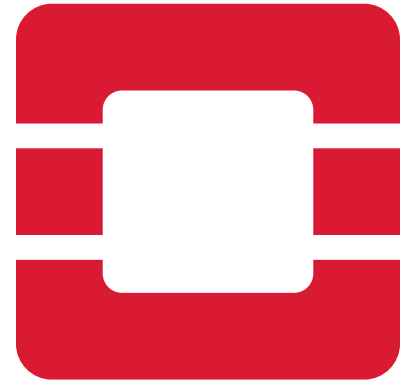
# 🌱 Pull images
kolla-ansible pull -i inventory/wrx

# 🚚 Upgrade
kolla-ansible upgrade -i inventory/wrx
```



# High Availability

- Keepalived & HAProxy (controllers)
- Stateless API Services
- MariaDB Galera Cluster as State Store
- Message Broker (RabbitMQ)
- Multiple nodes of each type



# Debugging Common

```
# Some useful commands for debugging

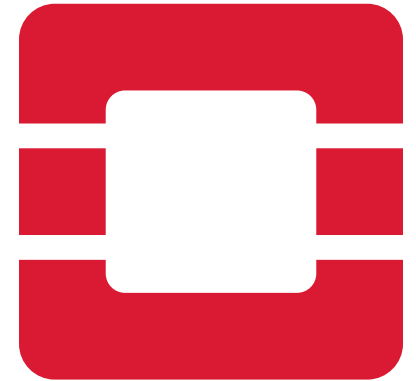
# On each node
docker ps | grep unhealthy
df -Ph

# On controller node
tail -f /var/log/kolla/mariadb/mariadb.log
tail -f /var/log/kolla/rabbitmq/rabbit*.log

# On deployment node
openstack compute service list
openstack volume service list
openstack network agent list

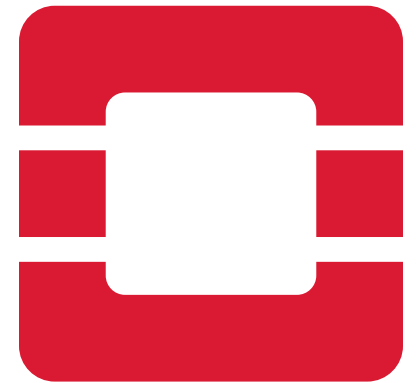
# Check HA IP on controller nodes
ip -br a

# Search for error messages connected with a resource
grep -r [RESOURCE_ID] /var/log/kolla/[SERVICE]
```



# Backup and Recovery

- MySQL Backup
- Config Backup (can be restored by kolla)
- Volumes Backup (can be ⚠ partially restored)
- Message Queues (are recreated if lost)



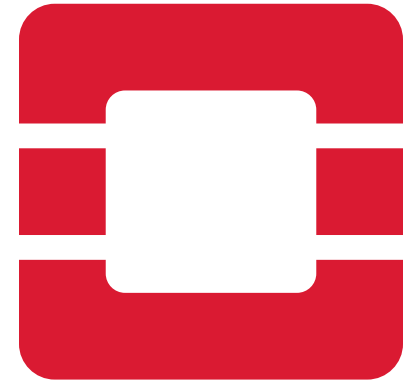
# Database Backup and Recovery

```
# Create a new backup
kolla-ansible mariadb-backup -i inventory/wrx

# Check on the controller
ls -lh /var/lib/docker/volumes/mariadb_backup/_data/

# To restore
gunzip -c <BACKUP_FILE> | docker exec -i mariadb mysql -u root -p

# If the DB cluster cannot create a quorum
kolla-ansible mariadb-recovery -i inventory/wrx
```



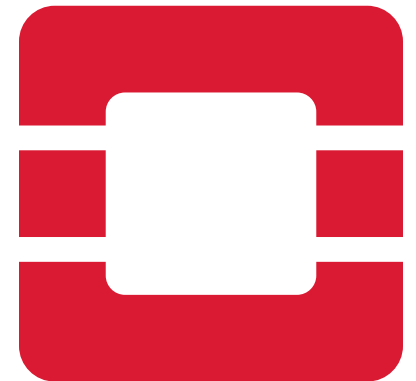
# Monitoring

- Ceilometer / Gnocchi / Aodh (for Billing)
- Prometheus / Alertmanager / Grafana

Grafana <http://int.os.wrx.sckt.net:3000>

Prometheus <http://int.os.wrx.sckt.net:9091>

Alertmanager <http://int.os.wrx.sckt.net:9095>

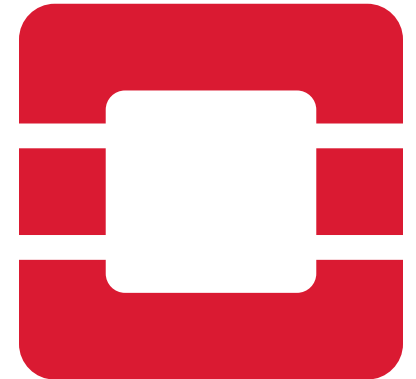


# Expand / Deploy a New Service

## Central Logging

```
# 🖱️ Enable central logging
cat custom-config/wrx/globals.d/logging.yml
sed -e 's/enable_central_logging: */enable_central_logging: "yes"/' \
    -i custom-config/wrx/globals.d/logging.yml
cat custom-config/wrx/globals.d/logging.yml

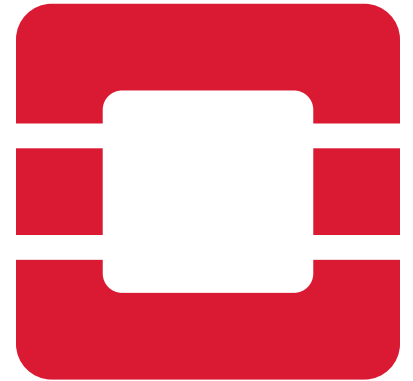
# 🚚 Deploy the new service
kolla-ansible pull          -i inventory/wrx --tags opensearch
kolla-ansible deploy        -i inventory/wrx --tags opensearch
kolla-ansible reconfigure   -i inventory/wrx --tags common
```



Visit:

- <http://int.os.wrx.sckt.net:5601>
- visualize, create an index pattern
- discover

# Using OpenStack



# Images

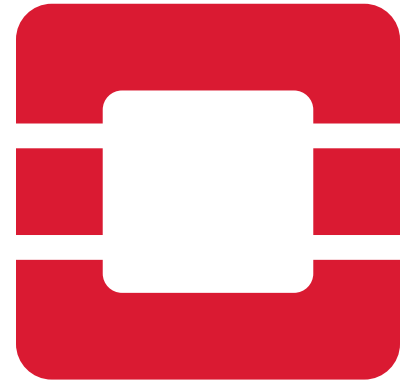
```
# Prepare images
mkdir -p ~/cloud-images/

ALMA_VERSION="10.2"
ALMA_IMG_NAME=$(
  curl -sS https://repo.almalinux.org/almalinux/10/cloud/x86_64/images/CHECKSUM \
    | grep GenericCloud-$ALMA_VERSION | awk '{print $2}' | sort | tail -n1)
ALMA_LINK=https://repo.almalinux.org/almalinux/10/cloud/x86_64/images/$ALMA_IMG_NAME

ROCKY_VERSION="10.2"
ROCKY_IMG_NAME=$(
  curl -sS https://dl.rockylinux.org/pub/rocky/10/images/x86_64/CHECKSUM \
    | grep SHA256 | awk '{print $2}' | grep GenericCloud-LVM-$ROCKY_VERSION \
    | sed -e "s/[()]/ /g" | sort | tail -n1)
ROCKY_LINK=https://dl.rockylinux.org/pub/rocky/10/images/x86_64/$ROCKY_IMG_NAME

UBUNTU_VERSION="24.04"
UBUNTU_IMG_NAME=noble-server-cloudimg-amd64.img
UBUNTU_LINK="https://cloud-images.ubuntu.com/noble/current/$UBUNTU_IMG_NAME"

DEBIAN_VERSION="13"
DEBIAN_IMG_NAME=debian-13-generic-amd64.qcow2
DEBIAN_LINK="https://cloud.debian.org/images/cloud/trixie/latest/$DEBIAN_IMG_NAME"
```



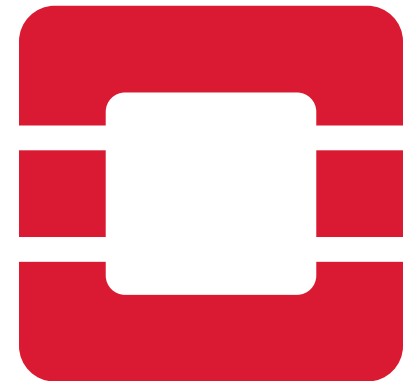
# Images

```
# Download images
for i in alma~$ALMA_VERSION~$ALMA_LINK \
        rocky~$ROCKY_VERSION~$ROCKY_LINK \
        ubuntu~$UBUNTU_VERSION~$UBUNTU_LINK \
        debian~$DEBIAN_VERSION~$DEBIAN_LINK
do
    IFS="~" read -r -a fields <<< $i
    name=${fields[0]}
    version=${fields[1]}
    link=${fields[2]}
    filepath=~/.cloud-images/${basename $link}

    echo "Downloading $name-$version image"
    curl --output-dir ~/.cloud-images -L -O $link

    echo "Uploading $name-$version image into OpenStack"
    openstack image create "$name-$version" \
        --file "$filepath" \
        --disk-format qcow2 \
        --container-format bare \
        --property os_distro="$name" \
        --property os_version="$version" \
        --public
done

# Check
openstack image list --long
```



# Authentication / Authorization

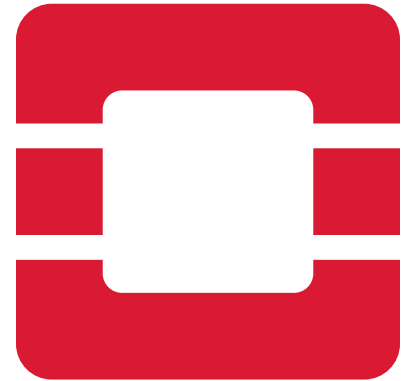
```
# Set permissions
PRJ_NAME=workshop
USR_NAME=workshop
USR_PASS=workshop

# Create project
openstack project create                $PRJ_NAME
openstack quota set --instances 32 --cores 64 $PRJ_NAME

# Create user
openstack user create --password $USR_PASS --project $PRJ_NAME $USR_NAME

# Create permissions for the user
openstack role add --user $USR_NAME --project $PRJ_NAME member

# Check
openstack role assignment list --name
openstack quota show $PRJ_NAME
```



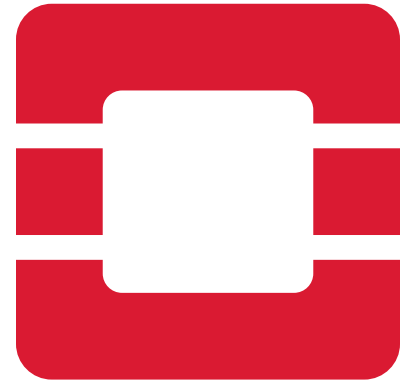
# How to authenticate with CLI

```
# Prepare an openrc file to be used with the OpenStack CLI
cp custom-config/wrx/admin-openrc.sh custom-config/wrx/workshop-openrc.sh

sed -e "s/OS_PROJECT_NAME=.* /OS_PROJECT_NAME=$PRJ_NAME/" -i custom-config/wrx/workshop-openrc.sh
sed -e "s/OS_TENANT_NAME=.* /OS_TENANT_NAME=$PRJ_NAME/" -i custom-config/wrx/workshop-openrc.sh
sed -e "s/OS_PASSWORD=.* /OS_PASSWORD=$USR_PASS/" -i custom-config/wrx/workshop-openrc.sh
sed -e "s/OS_USERNAME=.* /OS_USERNAME=$USR_NAME/" -i custom-config/wrx/workshop-openrc.sh

# Review
cat custom-config/wrx/workshop-openrc.sh
# Activate
source custom-config/wrx/workshop-openrc.sh

openstack network list
openstack router list
openstack server list
```



# Networking

- Networks:
  - flat
  - vlan
  - vxlan
- Routers
- Floating IPs
- Load Balancers

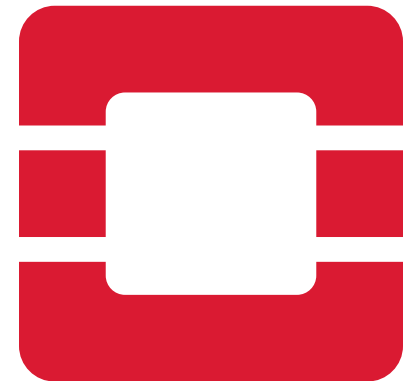
# Networking

## External Network

```
# Create external network (VLAN backed)
# as admin
EXT_NET_VLAN=3010
EXT_NET_NAME="shared${EXT_NET_VLAN}"
EXT_NET_CIDR='10.30.10.0/24'
EXT_NET_RANGE='start=10.30.10.192,end=10.30.10.254'
EXT_NET_GATEWAY='10.30.10.1'

openstack network create      ${EXT_NET_NAME} \
  --provider-physical-network physnet3      \
  --provider-network-type      vlan          \
  --provider-segment           ${EXT_NET_VLAN} \
  --external

openstack subnet create      ${EXT_NET_NAME}-subnet \
  --allocation-pool          ${EXT_NET_RANGE}      \
  --network                   ${EXT_NET_NAME}      \
  --subnet-range             ${EXT_NET_CIDR}       \
  --gateway                   ${EXT_NET_GATEWAY}   \
  --no-dhcp
```



# Networking

## Project Network

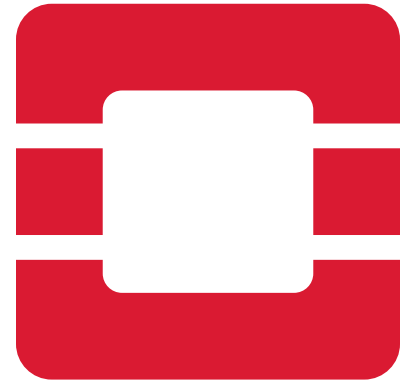
```
# Create project network (VXLAN backed)
# as workshop user in workshop project

# Activate
source custom-config/wrx/workshop-openrc.sh

PRJ_NAME=workshop
openstack network create workshop-net

openstack subnet create workshop-subnet \
  --subnet-range 10.10.10.0/24          \
  --network workshop-net                \
  --gateway 10.10.10.1

openstack router create workshop-router
openstack router add subnet workshop-router workshop-subnet
openstack router set --external-gateway shared3010 workshop-router
```



# Networking

## Virtual Machines

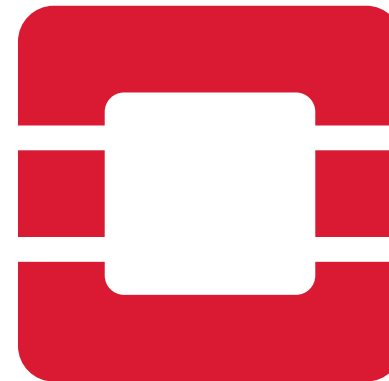
```
# as workshop user in workshop project

IMAGE_NAME=ubuntu-24.04
# IMAGE_NAME=alma-10.1
# IMAGE_NAME=rocky-10.1
# IMAGE_NAME=debian-13

openstack keypair create --public-key ~/.ssh/id_ecdsa.pub mykey

openstack server create          \
  --image      ${IMAGE_NAME}      \
  --flavor     m1.small            \
  --key-name   mykey              \
  --nic        net-id=workshop-net \
  vm0

openstack server create          \
  --image      ${IMAGE_NAME}      \
  --flavor     m1.small            \
  --key-name    mykey              \
  --nic         net-id=workshop-net \
  --boot-from-volume 10           \
  vm1
```



# Networking

## How to access?

```
# Create a config for a VLAN interface
# on the testing node as root
myid=$(cat /etc/env.json | jq .id)

cat > /etc/netplan/80-vlan.yaml <<EOF
network:
  version: 2
  vlans:
    xtrn0.3010:
      addresses:
        - "10.30.10.${myid}/24"
      dhcp4: false
      dhcp6: false
      id: 3010
      link: eth3
      link-local: []
EOF

chmod 600 /etc/netplan/80-vlan.yaml

netplan apply
```



# Networking

## How to access?

```
# Copy the ssh key from deployment to testing node
scp ~/.ssh/id_ecdsa* testing.mgmt.wrx.sckt.net:~/.ssh/

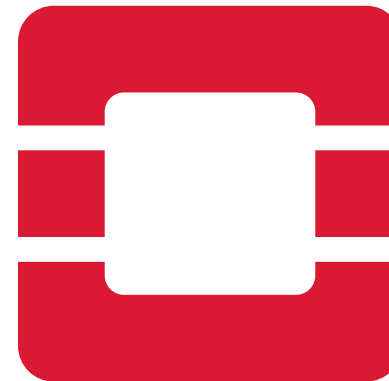
# Assign floating IPs to VMs
openstack floating ip create shared3010
FIP_VM0=$(
  openstack floating ip create shared3010 -f value -c floating_ip_address
)
openstack server add floating ip vm0 $FIP_VM0

FIP_VM1=$(
  openstack floating ip create shared3010 -f value -c floating_ip_address
)
openstack server add floating ip vm1 $FIP_VM1

# Review assigned IPs
openstack server list

# Allow access to vm0
openstack security group create allow-simple-access
openstack security group rule create allow-simple-access \
  --protocol tcp \
  --dst-port 22 \
  --remote-ip 0.0.0.0/0 \
  --ingress
openstack security group rule create allow-simple-access \
  --protocol icmp \
  --remote-ip 0.0.0.0/0 \
  --ingress
openstack server add security group vm0 allow-simple-access

# You should be able to connect to VMs from the testing node
ping -c 3 <IP>
ssh -i ~/.ssh/id_ecdsa ubuntu@<IP>
```



# Networking

## Debugging

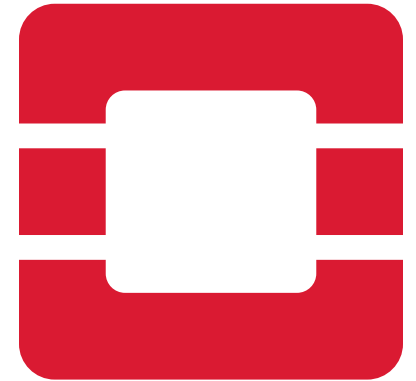
```
# Check VMs and IPs assigned to them
openstack server list --long

# Check the state of network services
openstack network agent list

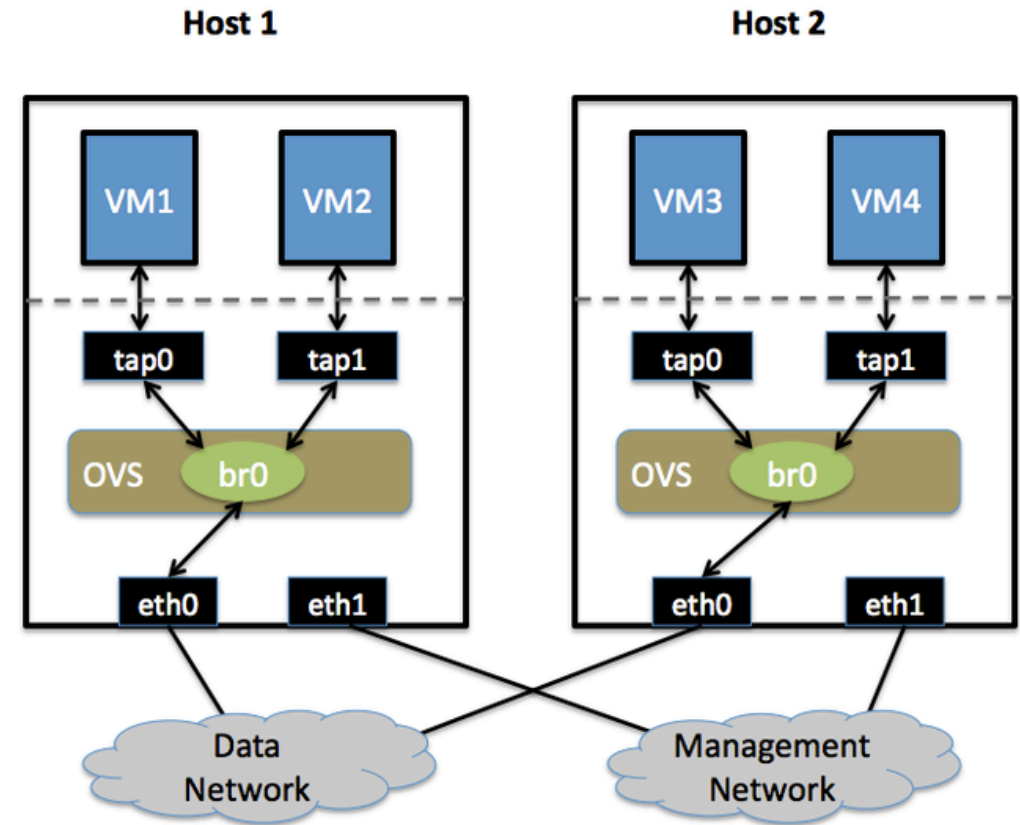
# Check the router has connectivity to the external network
# Check the router has IPs from the tenant network assigned
openstack router show workshop-router

# On network nodes check if there are network namespaces created for the router
# qrouter-<ROUTER_ID>
# snat-<ROUTER_ID>
# fip-<FLOATING_IP_NETWORK_ID>
ip netns
# Check if IP addresses are assigned
ip netns exec <NETWORK_NAMESPACE_ID> ip a

# On compute nodes check if there are network namespaces created for the router
ip netns
```

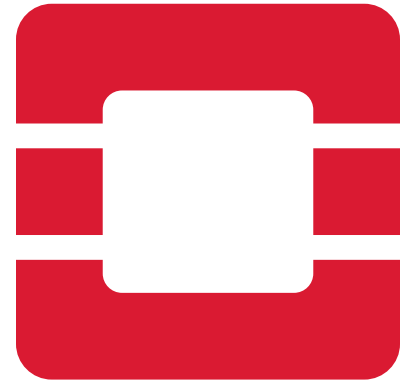


# Network Namespaces



# User Config with Cloud-Init

```
# Create user-data.yaml
cat > user-data.yaml <<EOF
#cloud-config
runcmd:
  - mkdir -p /var/lib/data/test
  - echo "<h1>Hello from write_files</h1>" > /var/lib/data/test/hello.html
fs_setup:
  - device:      /dev/vdb
    filesystem:  ext4
    overwrite:   true
mounts:
  - [ /dev/vdb, /var/lib/data, "ext4", "defaults,nofail", "0", "2" ]
write_files:
  - path:        /var/lib/data/index.html
    owner:       root:root
    permissions: '0644'
    content: |
      <h1>Hello from write_files</h1>
EOF
```



# User Config with Cloud-Init

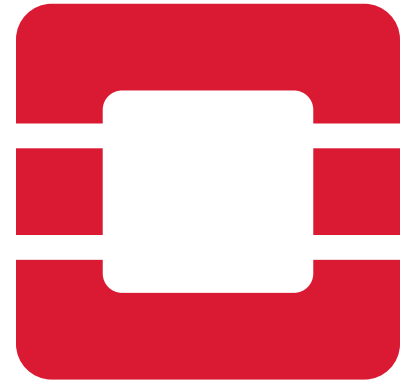
```
# Create a new VM with an additional volume
root_uuid=$(
  openstack volume create \
    --image "${IMAGE_NAME}" --size 10 -c Id -f value vm2-root
)
data_uuid=$(openstack volume create --size 10 -c Id -f value vm2-data )

openstack volume list

openstack server create \
  --flavor m1.small \
  --key-name mykey \
  --nic net-id=workshop-net \
  --block-device uuid=$root_uuid,source_type=volume,destination_type=volume,boot_index=0 \
  --block-device uuid=$data_uuid,source_type=volume,destination_type=volume,boot_index=-1 \
  --user-data ./user-data.yaml \
  vm2

# Inspect
openstack server show vm2 -fyaml
openstack volume list

# Create a new floating IP and attach it to the server
vm2_ip=$(openstack floating ip create -f value -c floating_ip_address shared3010)
openstack server add floating ip vm2 $vm2_ip
```

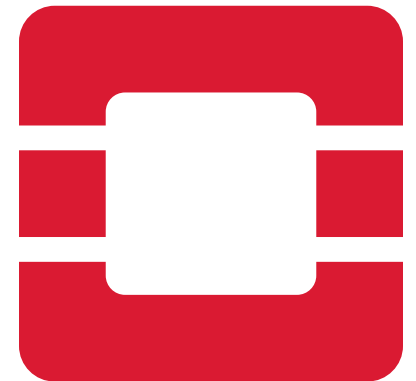


# User Config - Inspect the Created VM

```
# Log in from testing node
ssh -i ~/.ssh/id_ecdsa ubuntu@<FLOATING_IP>

cat /etc/fstab
mount | grep vdb

cat /var/lib/data/index.html
cat /var/lib/data/test/hello.html
```



# Debugging

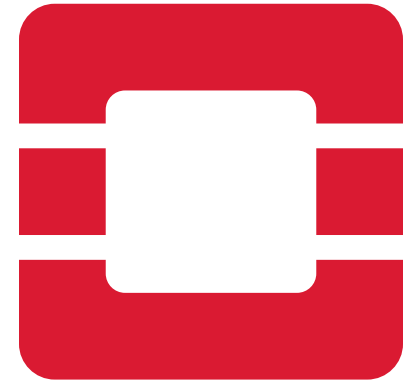
```
# Inspect the instance
openstack server show vm2

# Inspect VM with libvirt on the compute host
docker exec -ti nova_libvirt virsh list
docker exec -ti nova_libvirt virsh dumpxml <INSTANCE_NAME>
docker exec -ti nova_libvirt virsh console <INSTANCE_NAME> # ctrl+5 to exit
docker exec -ti nova_libvirt virsh domblklist <INSTANCE_NAME>
docker exec -ti nova_libvirt virsh domiflist <INSTANCE_NAME>

# Inspect volumes backed by Ceph on the compute host
docker exec -ti nova_compute ls /etc/ceph/
docker exec -ti nova_compute ceph -s
docker exec -ti nova_compute ceph --id cinder -s
docker exec -ti nova_compute ceph --id cinder osd pool ls
docker exec -ti nova_compute rbd --id cinder ls images
docker exec -ti nova_compute rbd --id cinder ls vms
docker exec -ti nova_compute rbd --id cinder ls volumes

# Inspect network interface on the compute host
docker exec -ti openvswitch_vswitchd ovs-vsctl show

# Within the instance
cloud-init status
cloud-init status --long
```



# Block Devices

```
# How many VMs are running? How many volumes should we see?
openstack volume list

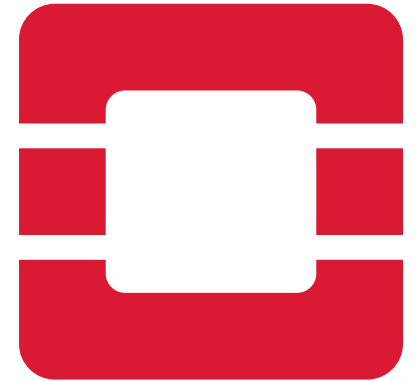
# Create a new volume and attach it
openstack volume create --size 1 vm1-data
openstack server add volume vm1 vm1-data
openstack volume attachment list

# Inspect it within the VM from the testing node
lsblk

# Create some data on it
mkfs.ext4 /dev/vdb
echo "$(date) Hello from vm1"

# Take a snapshot on the deployment node
openstack volume snapshot create --volume vm1-data vm1-data-snap1 [--force]
openstack volume snapshot list

# Change the data
echo "Let us add a new line into the file after the snapshot was created" >> /mnt/test
cat /mnt/test
```

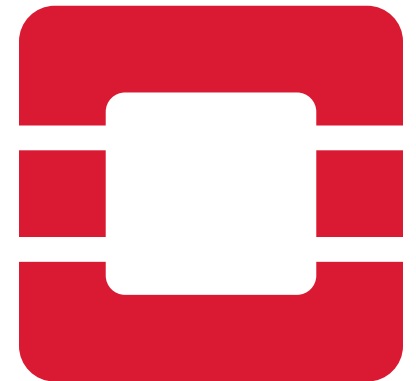


# Block Devices

## Revert a Volume

```
# Restore from the snapshot
openstack server stop    vm1
openstack server list; openstack volume list;
openstack volume snapshot list --volume vm1-data

openstack server remove volume vm1 vm1-data
openstack volume revert          vm1-data-snap1
openstack server add    volume vm1 vm1-data
openstack server start          vm1
```



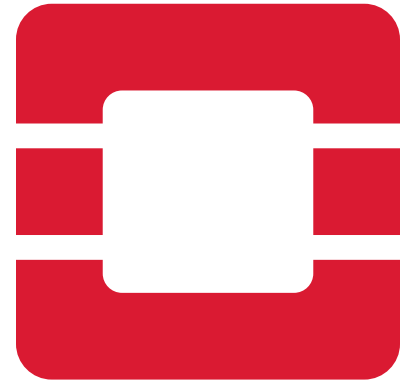
# Block Devices

## More Operations

```
# Create a volume from an image
openstack volume create --image cirros --size 1 vm1-cirros

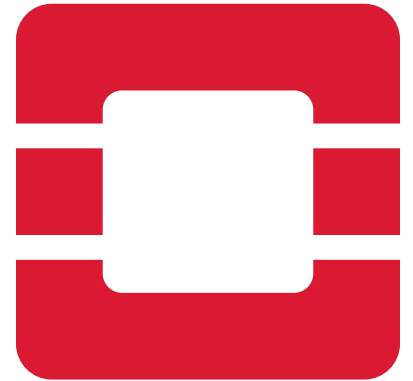
# Create a volume from a snapshot
openstack volume snapshot create --volume vm2-root vm2-root-snap1 [--force]
openstack volume create --snapshot vm2-root-snap1 ubuntu-24.04-from-vm2

# Create an image from a volume
openstack image create --volume ubuntu-24.04-from-vm2 ubuntu-24.04-custom
openstack image list
```



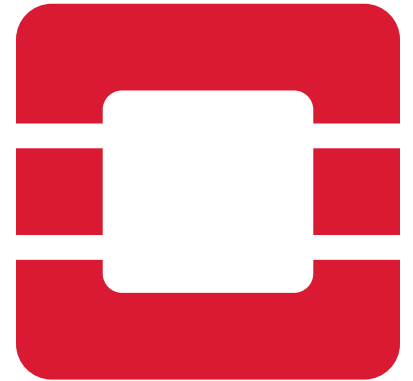
# Resource Segregation

- Projects
- Host Aggregates
- Volume Backends
- Network Agents
- Availability Zones



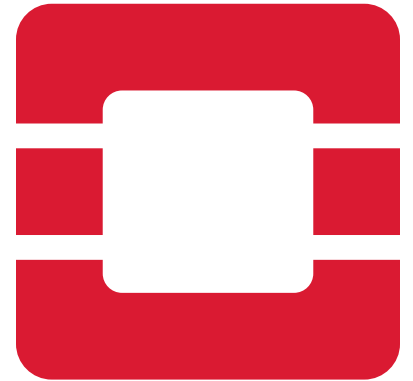
# Availability Zones

is a human-readable label that represents a **physically isolated slice of data center infrastructure**. It is the primary tool used to group hardware into separate **fault domains** so that a single physical failure does not take down an entire cloud application.



# Availability Zones

- Power Distribution Unit
- Data Center (Room)
- Cooling Domain
- Top-of-Rack (ToR) Switch Domain
- Storage Backend



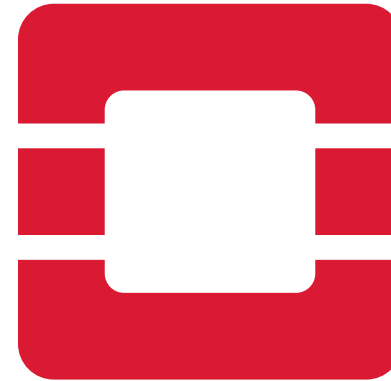
# Availability Zones

```
openstack aggregate create --zone az1 prd-az1
openstack aggregate create --zone az2 prd-az2

openstack aggregate add host prd-az1 compute01
openstack aggregate add host prd-az1 compute02
openstack aggregate add host prd-az1 compute03

openstack aggregate show prd-az1
openstack aggregate show prd-az2

openstack availability zone list --compute
openstack availability zone list --network
openstack availability zone list --volume
```



# Host Aggregates

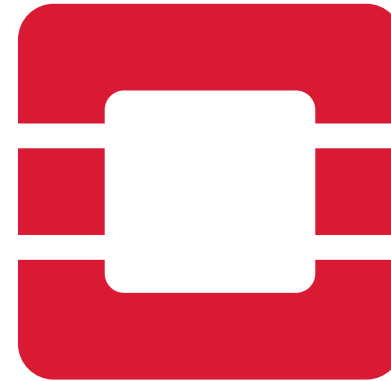
```
openstack aggregate create hw-type-A
openstack aggregate create hw-type-B
openstack aggregate create windows
openstack aggregate create gpu

openstack aggregate add host hw-type-A compute01
openstack aggregate add host hw-type-A compute02
openstack aggregate add host hw-type-A compute03
openstack aggregate add host hw-type-B compute11

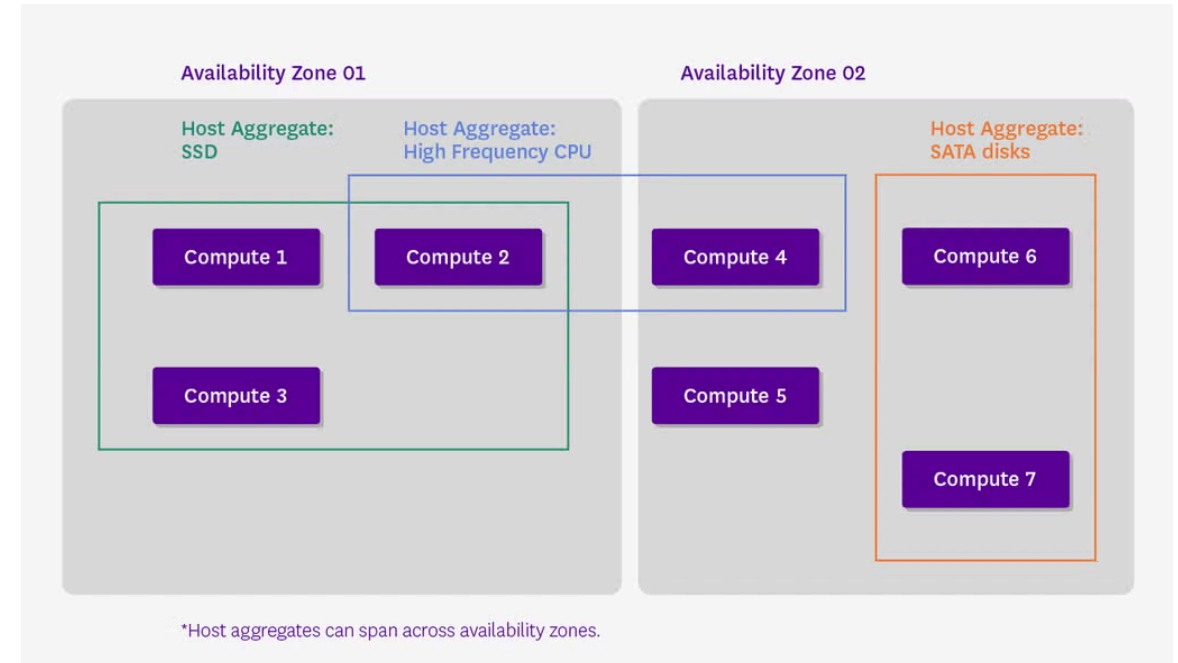
openstack aggregate add host windows compute01
openstack aggregate add host windows compute11

openstack aggregate add host gpu compute11

openstack aggregate show windows
```



# Host Aggregates vs. Availability Zones



# Links

- [OpenStack - Docs](#)
- [OpenStack - Configuration Guides](#)
- [OpenStack - Releases](#)
  
- [OpenStack - Horizon / Dashboard](#)
- [OpenStack - Skyline / Dashboard](#)
- [OpenStack - Grafana / Monitoring](#)
- [OpenStack - Prometheus / Monitoring](#)
- [OpenStack - Alertmanager](#)

# Links

- [Cloud-Init - Docs](#)
- [Harbor - Registry](#)

# Feedback and contributions

Found an error or want to contribute? Open a PR on [GitHub](#).

More labs and cheat sheets: [infraguide.org](https://infraguide.org)

Questions or ideas: [ping@infraguide.org](mailto:ping@infraguide.org)

